TT-I-00549C
May 14, 1974
SUPERSEDING
Int. Fed. Spec. TT-I-00549B
June 5, 1959
INTERIM REVISION OF
Fed. Spec. TT-I-549A
October 21, 1948

INTERIM FEDERAL SPECIFICATION

INK, WRITING, COLORED (NON-PERMANENT)

This Interim Federal Specification was developed by the Federal Supply Service, General Services Administration, Washington, DC 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to Federal Specification TT-I-549a dated October 21, 1948.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers writing ink intended for general use. It is not water resistant and is not suitable for writing permanent records.

1.2 Classification.

1.2.1 Type. The ink shall be of one type.

1.2.2 Colors. The ink shall be furnished in the following colors, as specified in the invitation for bids, contract, or order (see 3.2 and 6.1): Red, green, purple, and royal blue.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:
O-G-491 - Glycerol, Technical (High Gravity)
UU-P-121 - Paper, Bond and Writing; White and Colored
PPP-B-636 - Boxes, Shipping, Fiberboard

Federal Standards:
Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies)
Fed. Std. No. 191 - Textile Test Methods

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 7510

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

The Pharmacopoeia of the United States.

(The Pharmacopoeia of the United States is published by Mack Publishing Company, 20th and Northampton Streets, Easton, Pennsylvania 18042).

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section. 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Material. The manufacturer is not restricted in a choice of dyes for making the ink covered by this specification. The quality of the ink shall be at least equal to that for the colors described in 3.2.

3.2 Quality and color. The ink as received shall be a clear solution not inferior in any respect to one prepared by dissolving the required amount of dye for the respective colors, as specified below, in 1 liter of distilled water containing at least 6 percent of glycerol conforming to Federal Specification O-G-491 and 1 gram (gm) of Phenol or 1 gm. of Sodium Benzoate as prescribed in the Pharmacopoeia of the United States.

Red: 5.5 gm. of Crocein Scarlet (British Colour Index No. 252), or 5.2 gm. of azophloxine (British Colour Index No. 31).

Green: 7.5 gm. of Alkali Fast Green, 300 percent concentrate, (British Colour Index No. 735).

Purple: 21.5 gm. of Acid Wool Violet (British Colour Index No. 698).

Royal Blue: 10.0 gm. of Soluble Blue (British Colour Index No. 707).

3.3 Properties. The sample shall be equal to the comparison ink (par. 3.2) with respect to the following requirements: color of streaks when freshly made, permanence to light, and freedom from sediment before and after exposure to light and air. Writing produced by the ink shall photograph clearly on microcopying film. (see 4.4.5).

3.4 Containers (bottles). Unless otherwise specified, the ink shall be furnished in 2-ounce, 4-ounce, 1-pint, or 1-quart bottles, as ordered (see section 5 and 6.1).

3.5 Workmanship. The ink shall be manufactured in accordance with good commercial practice. The commodity shall be furnished free from dirt or any other foreign matter, which may impair its utility.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. For purposes of inspection and testing, a lot shall consist of all units offered for delivery at one time.

4.3 Sampling.

4.3.1 For lot acceptance inspection. Samples of ink for acceptance inspection and testing shall be selected from each production lot in accordance with Military Standard MIL-STD-105, Inspection Level II.

4.3.2 Samples. A sample shall consist of an original unopened bottle of ink bearing all the manufacturer's marks when the bottle contains not less than 4 fluid ounces. When the ink is furnished in smaller bottles, enough of them to aggregate at least 4 fluid ounces shall be submitted unopened and with all marks intact. The contents of these bottles shall be mixed to form a composite sample.

4.4 Inspection.

4.4.1 Acceptance inspection. The sample ink of any color shall be allowed to remain undisturbed in a stoppered glass container for 24 hours at temperatures in the range of 21° to 27°C. to allow any sediment to settle. The container shall then be held against the light and slowly tilted away from the observer who can then see any sediment. A sufficient quantity of the clear ink shall be removed with a pipette for conducting the tests specified in 4.5. Failure to meet any one or more of these tests shall be cause for rejection of the entire lot offered for delivery.

4.4.2 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements in section 5 of this specification. Defects shall be scored in accordance with table I. For examination of interior packaging, the sample unit shall be one shipping container fully prepared for delivery, selected at random just prior to the closing operations. Sampling shall be in accordance with MIL-STD-105. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 with an acceptable quality level (AQL) of 4.0 defects per hundred units.

TABLE I. Classification of preparation for delivery defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, or distortion of container.
Content (exterior and interior container)	Number per container is more or less than required. Net weight exceeds requirements.

4.5 Tests. The sample shall be tested as follows in comparison with the applicable ink made as described in 3.2.

4.5.1 Ink streaks. Streaks shall be made by allowing measured portions of the ink, about 0.6 milliliter each, to flow freely across a sheet of white bond paper conforming to UU-P-121, type III, 40-pound, which is pinned to a board or clamped to a pane of glass and held at an inclination of 45°. A streak of the ink undergoing test shall be alternated with a streak of the comparison ink on the same sheet of paper and four streaks of each ink shall be made in the machine direction of the paper. (see 6.2 and 6.3).

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4.5.2 Form, color, and penetration. When the streaks are dry, the sheets shall be examined on the front and on the reverse sides. The streaks of the sample shall have the same general form and shall have approximately the same hue and intensity of color as those of the comparison ink. When viewed from the front and from the back, the streaks of the sample and those of the comparison ink shall be equally uniform in color and in degree of penetration.

4.5.3 Resistance to light. The paper shall be cut into strips 1 inch in width at right angles to the streaks. After the ink has been allowed to dry for at least 1 hour, the strips shall be used to make the following test: After exposure for 24 hours to the radiation from a glass-enclosed carbon arc such as the Fade-Ometer described in Fed. Std. No. 191, the streaks of the sample shall show no more evidence of fading than those of the comparison ink.

4.5.4 Stability. Twenty-five milliliters of the sample and 25 ml. of the comparison ink shall each be allowed to stand undisturbed in separate but similar colorless glass crystallizing dishes of 2-inch inside diameter and about 1-3/4 inch depth, loosely covered to exclude dust. After 7-days exposure to diffused daylight and air at ordinary room temperature, the sample ink shall be as free from sediment and mold as the comparison ink.

4.5.5 Reproducibility on microcopying film. At least 10 lines consisting of all the letters of the alphabet shall be made with a suitable writing pen on type III, 40-pound, white bond paper conforming to Federal Specification UU-P-121, with the ink under test. A similar number of lines shall be made with the comparison ink on a separate sheet of the same type bond paper. Both sheets of paper then shall be photographed with microcopying film. The reproduction of the sample ink with microcopying film. The reproduction of the sample ink shall be as clear and legible as that of the comparison ink.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. The volume of ink specified (i.e., 2-ounces, 4-ounces, 1-pint, or 1-quart) shall be packaged in commercial-type bottles in accordance with the manufacturer's standard practice. No unit (individual) boxing is required for 2-ounce or 4-ounce bottles. One-pint and one-quart shall be unit (individually) packaged in fiberboard boxes conforming to PPP-B-636, type CF, class weather resistant, style RSC, sized to snugly fit the bottle contained. Both ends of the box shall be closed in accordance with method V of the box specification to prevent accidental opening (see 5.3.1.2)

5.1.1.2 Intermediate packing. Twelve (12) each unit bottles of either the 2-ounce or 4-ounce size, shall be intermediately packaged in a fiberboard box of appropriate size conforming to PPP-B-636, type CF, class weather resistant, style RSC, fitted with a full or shoulder-height corrugated fiberboard partition assembly providing separate cells or sections for each bottle. Fiberboard for the box shall be closed in accordance with method V of the appendix to the box specification. No intermediate packaging is required for 1-pint and 1-quart bottles packaged in unit box as specified in 5.1.1.1.

5.1.2 Level B.

5.1.2.1 Unit packaging. The volume of ink specified (i.e., 2-ounces, 4-ounces, 1-pint, or 1-quart) shall be packaged in commercial-type bottles in accordance with the manufacturer's standard practice. No unit (individual) boxing is required for 2-ounce or 4-ounce bottles. One-pint and one-quart bottles shall be unit (individually) packaged in corrugated fiberboard boxes, double-slide type, sized to snugly fit the bottle contained. Both ends of the box shall be tape-sealed to prevent accidental opening (see 5.3.1.2).

5.1.2.2 Intermediate packaging. Twelve (12) each unit bottles of either the 2-ounce or 4-ounce size, shall be intermediately packaged in a fiberboard box of appropriate size conforming to PPP-B-636, type CF, class domestic, fitted with full- or shoulder-height corrugated fiberboard partition assembly providing separate cells or sections for each unit bottle. Fiberboard for this box shall test not less than 175 p.s.i., and all flaps shall be secured by use of suitable adhesive or stitching or a combination of the two methods. No intermediate packaging is required for 1-pint and 1-quart bottles packaged in unit boxes as specified in 5.1.2.1.

5.1.3 Level C. The ink shall be packaged in accordance with the manufacturer's standard practice.

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5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Four (4) intermediate boxes 12 unit containers each (48 bottles) of either 2-ounce of 4-ounce size, shall be packed in a close-fitting shipping container. Twelve (12) unit boxes (12 bottles) of either 1-pint or 1-quart size, shall be packed in a close-fitting container. Shipping container shall conform to PPP-B-636, type CF or SF, class weather-resistant, style RSC or CSSC, of fiberboard testing not less than 200 p.s.i. The box shall be closed in accordance with method V of the appendix to the box specification.

5.2.2 Level B. Four (4) intermediate boxes of 12 unit containers each (48 bottles) of either 2-ounce or 4-ounce size, shall be packed in a close-fitting shipping container. Twelve (12) unit boxes (12 bottles) of either 1-pint or 1-quart size, shall be packed in a close-fitting shipping container. Shipping containers shall conform to PPP-B-636, type CF or SF, class domestic, style RSC or CSSC, of fiberboard testing not less than 200 p.s.i. The box shall be closed in accordance with method II of the appendix to the box specification.

5.2.3 Level C. The ink packaged as specified in 5.1 shall be packed to comply with the National Motor Freight Classification rules or Uniform Freight Classification rules, as applicable.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, all the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.1.1 Unit containers (bottles). Unless otherwise specified each bottle shall be labeled with the name of the commodity, applicable Federal Stock Number, composition, color, volume of content, and name of the manufacturer.

5.3.1.2 Unit packages and intermediate packages. Unless otherwise specified, unit and intermediate packages shall be marked or labeled on one side or end as appropriate, with the name of the commodity, applicable Federal Stock Number, composition, color, unit and quantity (e.g. Bottle -1-Pt.) or 12-Ea.-2-oz. Bottles) contract or order number, and name of manufacturer or contractor, if different. The word "TOP" with arrows pointing to the top shall appear on at least two (2) sides of the unit boxes to indicate the position of the 1-pint or 1-quart bottle within.

5.3.2 Military agencies. In addition to markings required by contract or order, all the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options offered herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Color of ink required (see 1.2.2).
(c) Size of container (see 3.4).
(d) Data; inspection responsibility if other than as specified herein (see 4.1).
(e) Levels of packaging and packing and any special marking for shipment required (see 5.1, 5.2, and 5.3).
(f) Whether civil or military marking is required (see 5.3).

6.2 For making ink streaks (par. 4.3.1), a piece of glass tubing about 3.5 millimeters in bore and 250 millimeters long is used. A mark etched or scratched 62 millimeters from one end indicates the required volume, 0.6 milliliter, of ink. The ends of the tube can be fire polished, but should not be constricted. By drawing the ink up to the mark, holding the tube vertically against the paper near the top, and allowing the ink to flow suddenly down across the paper, uniform streaks will be obtained. Separate clean, dry tubes should be provided for the standard and the sample.

6.3 When streaks are made (par. 4.3.1), the ink flows smoothly down the sheet when it runs in the machine direction of the paper. If it runs across the machine direction, or grain, the paper wraps into ridges which cause the ink to spread sideways and make uneven streaks. To determine the machine direction draw a line about 2 inches long on one corner of the sheet of paper. Cut a 1-inch square containing a portion of the line from the corner. Place the square on water and observe the curl. The parallel edges of the trough formed by the square as the paper curls away from the water indicate the machine direction.

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6.4 The dye numbers listed in paragraph 3.2 are from the Colour Index of the Society of Dyers and Colourists.

6.5 It is believed that this specification adequately describes the characteristics necessary to secure that desired commodity and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

COORDINATION WAIVED BY DOD

User Interest:

Army - GL

Preparing Activity:

GSA - FSS

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